

Synopsis

End-to-End Machine Learning Regression System for Institutional Energy Usage Prediction

Problem Id: 23

Student Names : Vinamra Semwal, Mayank Fikwal

University Roll numbers : 2028907, 2028764

Section – K

• Problem Description:

The objective of this project is to design and develop an end-to-end machine learning regression system to analyze and predict energy usage in institutional settings such as universities, colleges, and corporate campuses. Institutions consume large and often unpredictable amounts of electricity driven by varying occupancy, weather conditions, and appliance usage — making manual energy management inefficient and costly. Predicting energy demand accurately can help in optimizing consumption, reducing operational costs, and contributing to sustainability goals.

The project uses a numerical dataset containing features such as time of day, number of students present, ambient temperature, and appliance usage. The data undergoes preprocessing including cleaning, normalization, and feature transformation to ensure quality and consistency. Exploratory Data Analysis is then performed through statistical summaries and visualizations like line plots, histograms, and correlation heatmaps to uncover consumption patterns, detect outliers, and understand relationships between features.

Feature selection techniques — PCA, Forward Selection, and Backward Elimination — are applied to retain only the most statistically significant predictors. Finally, regression models including Linear, Polynomial, and Ridge/Lasso regression are trained and evaluated using metrics such as MAE, RMSE, and R^2 , demonstrating a complete ML workflow from raw data to a deployable predictive model.

A machine learning regression model is then developed and evaluated to predict energy usage, demonstrating a complete ML workflow from raw data to evaluated predictive models. The processed dataset will serve as the foundation for model training and evaluation in subsequent phases.

• Steps of Implementation:

The implementation follows a structured machine learning pipeline consisting of the following phases:

1. **Data Collection:** Select a suitable numerical institutional energy dataset (e.g., UCI Energy Efficiency or custom campus data). Features include time of day, occupancy (number of students), ambient temperature, and appliance usage.
2. **Data Preprocessing:** Handle missing values, remove duplicate records, and encode categorical features. Apply normalization (MinMaxScaler / StandardScaler) and feature transformations (log, polynomial) for consistent data quality.
3. **Exploratory Data Analysis (EDA):** Compute statistical summaries (mean, std, quartiles). Generate visualizations including line plots for trends, histograms for feature distributions, and correlation heatmaps to detect multicollinearity.
4. **Feature Selection:** Apply PCA for dimensionality reduction. Use Forward Selection and Backward Elimination to retain only statistically significant features and reduce the risk of overfitting.
5. **Model Development:** Train regression models including Linear Regression, Polynomial Regression, and Ridge/Lasso on the cleaned, feature-selected dataset. Perform an 80/20 train-test split and apply k-fold cross-validation.
6. **Model Evaluation:** Evaluate models using MAE, MSE, RMSE, and R^2 score. Compare performance across models, plot actual vs. predicted values, and analyze residual plots to validate regression assumptions.
7. **Model Saving:** Serialize and save the best-performing trained regression model using Pickle or Joblib for later deployment and reuse.

• Prompt Used For Dataset Creation:

Generate 5000 row dataset csv file that creates a synthetic campus lab energy-usage dataset including timestamp, date, time-of-day hour, day of week, month, weekend flag, students present, ambient temperature, HVAC, lighting, computer and lab equipment loads as features, and total energy usage in kWh as a noisy function of these loads, with realistic daily and seasonal patterns and higher loads and occupancy in weekday daytime hours.

• Expected Modules / Libraries to be used:

- **pandas** – Data loading, cleaning, manipulation, and statistical summaries using DataFrames.
- **numpy** – Numerical computing; array operations and linear algebra for feature matrices.
- **matplotlib** – Creating line plots, histograms, and actual vs. predicted value plots.
- **seaborn** – Statistical visualizations including heatmaps, pair plots, and distribution plots.
- **scikit-learn** – Core ML library for preprocessing (scalers), feature selection (PCA, RFE), model training, and evaluation metrics (MAE, RMSE, R^2).
- **pickle / joblib** – Serializing and loading trained regression model artifacts.
- **Jupyter Notebook** – Interactive development environment for code, output, and markdown documentation.

• Expected Evaluation Metrics and Plots to be used:

Since the target variable is continuous (energy usage in kWh), regression-specific evaluation metrics will be used to properly assess model performance.

Evaluation Metrics:

1. **Mean Absolute Error (MAE):** Measures the average absolute difference between actual and predicted energy values. Easy to interpret in the original unit (kWh).
2. **Mean Squared Error (MSE):** Penalizes larger errors more heavily. Useful for detecting high-variance predictions.
3. **Root Mean Squared Error (RMSE):** Square root of MSE; provides error magnitude in the same unit as the target variable (kWh).
4. **R² Score (Coefficient of Determination):** Measures the proportion of variance in the target explained by the model. A value closer to 1.0 indicates a better fit.

Visualization Plots to be Used:

1. **Line Plot:** Visualize energy consumption trends over time to identify patterns and seasonal variation.
2. **Histogram:** Study the distribution of individual features such as temperature and energy usage.
3. **Correlation Heatmap:** Identify multicollinear features and select the most relevant predictors.
4. **Scatter Plot (Actual vs. Predicted):** Visually assess the accuracy of the trained regression model.
5. **Box Plot:** Detect outliers in each feature before and after the preprocessing step.

• Required Topics from the Subject:

1. **Python Fundamentals:** Data types, lists, dictionaries, and control flow form the backbone of data manipulation. Functions and lambda expressions are used extensively in EDA and preprocessing pipelines.
2. **NumPy Arrays and Matrix Operations:** Feature matrices are represented as 2D NumPy arrays. Operations such as matrix multiplication, transpose, and broadcasting are used in model training and PCA decomposition.
3. **File Handling and Data I/O:** Reading .csv datasets using Pandas (`pd.read_csv`), writing cleaned data, and saving model artifacts using Pickle or Joblib for later use.
4. **Statistics and Probability:** Descriptive statistics (mean, median, standard deviation), correlation coefficients, and probability distributions underlie EDA and metric interpretation.
5. **Regression Analysis:** Linear regression (ordinary least squares), polynomial feature expansion, and regularized regression (Ridge and Lasso) are the primary models developed in this project.
6. **Dimensionality Reduction and Feature Selection:** PCA reduces the high-dimensional feature space while preserving maximum variance. Forward and Backward Selection iteratively add or remove features based on statistical significance (p-value, AIC).
7. **Model Evaluation:** Evaluation metrics such as MAE, RMSE, and R² score are used to compare model performance and select the best regression model.

• Platform Required:

The project will be developed using Python programming language. Jupyter Notebook will be used as the primary environment for data analysis, visualization, and documentation. Visual Studio Code will be used for project development, pipeline implementation and debugging. Google Colab is available as an optional cloud-based alternative for resource-intensive computations. GitHub will be used for version control and project repository management.

• Books and Link Sources:

- **Scikit-learn Documentation** – Supervised Learning.
From: https://scikit-learn.org/stable/supervised_learning.html
- **Rahil Shaikh, "Feature Selection Techniques in Machine Learning with Python", *Towards Data Science*, Medium, October 28, 2018.**
Available at: <https://towardsdatascience.com/feature-selection-techniques-in-machine-learning-with-python>
- **Pandas Documentation** – Used for data cleaning, data manipulation, and exploratory data analysis (EDA).
From : <https://pandas.pydata.org/docs/>
- **Scikit-learn Documentation** – Used for data preprocessing techniques such as feature scaling, encoding categorical variables, and splitting the dataset into training and testing sets.
From : https://scikit-learn.org/stable/supervised_learning.html